

Question				Marking details	Marks available				Maths	Prac
					AO1	AO2	AO3	Total		
9	(a)	(i)		Use of intensity = $\frac{P}{4\pi R^2}$ or correct substitution (1) Intensity = 1 415 [W m ⁻²] (1)	1	1		2	2	
		(ii)		[Using 1 400 W m ⁻²]: 20% = 280 W m ⁻² 15% = 42 W m ⁻² (1) for both efficiency calculations Area = $\frac{8 \times 10^9}{42}$ (ecf) = 1.9 × 10 ⁸ m ² (1) (unit) Accept 190 km ² or equivalent		2		2	2	
		(iii)		Greater intensity of radiation in Portugal (accept increased/longer sunshine hours throughout year) Do not accept more efficient			1	1		
	(b)	(i)		$\lambda_{\max} T = \text{constant}$	1			1		
		(ii)		$\lambda_{\max} = \frac{0.0029}{290}$ (1) $\lambda_{\max} = 10 \times 10^{-6}$ [m] so in IR (1)		2		2	2	
		(iii)	I	Surface emits IR in region 5 μm to 20 μm (1) [Partially] absorbed by water vapour (and CO ₂) (1) Correct reference to numerical data from graphs e.g. either water: greatest absorption between 5 μm to 8 μm or 12 μm to 20 μm) or CO ₂ : (greatest absorption between 10 μm to 20 μm) (1) [IR] re-emitted in all directions (1)	1 1		1 1	4		
			II	Higher temperatures will lead to increased water vapour in atmosphere (1) Leading to further absorption and increase in temperature (1) Alternative: As the ice cap melts leaving darker surface (1) which absorbs more visible radiation and increase in temperature (1) OR Melting permafrost (1) means more methane in atmosphere (1)			2	2		

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	(c)	(i)		U value = Energy per second / (area $\times$ temperature difference) / rate of heat transfer per unit area per unit temperature difference / $U = \frac{P}{A\Delta\theta}$ with symbols correctly identified (1) Thickness of material or conductivity of material (1)	2			2		
		(ii)		Energy per second lost = $1.8 \times (5 \times 3) \times (21 - 9)$ (1) for substitution Energy per second = 324 [W] (1)		2		2	2	
		(iii)		$1\,000 = 1.8 \times 15 \times (21 - T)$ (1) $T = -16\text{ }^{\circ}\text{C}$ (1)		2		2	2	
				Question 9 total	6	9	5	20	10	0